

Name: _____ Date: _____

Period: _____

Primary Objective

I will be able to...

- LO 3.6.A** — Identify the amplitude, vertical shift (midline), period, and phase shift of a sinusoidal function from its equation. _____
- LO 3.6.A** — Explain how each parameter in $f(\theta) = a \sin(b(\theta + c)) + d$ corresponds to an additive or multiplicative transformation of $y = \sin \theta$. _____

Vocabulary & Key Concepts

sinusoidal function: A function equivalent to $a \sin(b(\theta + c)) + d$ (or the cosine form) with $a \neq 0$. A _____ of the parent sine or cosine function.

amplitude: Equals $|a|$ — half the vertical distance from the _____ to the _____. Controls the vertical _____.

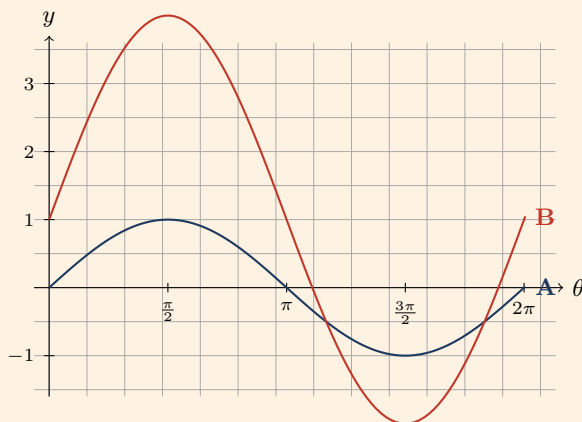
midline: The horizontal line $y = \underline{\hspace{1cm}}$. The vertical _____ of the function. Average of max and min outputs.

period: Length of one complete cycle; $T = \frac{2\pi}{\underline{\hspace{1cm}}}$. Controlled by the horizontal dilation factor b .

phase shift: Horizontal translation equal to _____ when the function is written in factored form $b(\theta + c)$. Positive c shifts _____; negative c shifts _____.

Hook

Entry Question: The two sinusoidal functions below are shown on the same axes. Use the graphs to answer the questions.



- Q1.** Graph A has amplitude _____ and midline $y = \underline{\hspace{1cm}}$.
- Q2.** Graph B has amplitude _____ and midline $y = \underline{\hspace{1cm}}$.
- Q3.** What equation could describe Graph B? $y = \underline{\hspace{2cm}}$

LO 3.6.A — Part 1: Vertical Transformations (a and d)

Vertical Dilation: $g(\theta) = a \sin(\theta)$

Starting from $y = \sin \theta$, multiplying all outputs by a changes the _____. The new amplitude is _____. If $a < 0$, the graph is also _____ over the midline.

Function	Amplitude	Max	Min	Midline
$y = \sin \theta$				
$y = 4 \sin \theta$				
$y = -2 \sin \theta$				
$y = \frac{1}{2} \sin \theta$				

Vertical Translation: $g(\theta) = \sin(\theta) + d$

Adding d to all outputs shifts the _____ to $y = d$. The amplitude is _____ (unchanged).

Function	Amplitude	Max	Min	Midline
$y = \sin \theta + 3$				
$y = 2 \sin \theta - 1$				

LO 3.6.A — Part 2: Horizontal Transformations (b and c)

Horizontal Dilation: $g(\theta) = \sin(b\theta)$

Replacing θ with $b\theta$ compresses or stretches the graph horizontally. New period = $\frac{2\pi}{|b|}$.

If $|b| > 1$: the graph is _____ (period is shorter).

If $|b| < 1$: the graph is _____ (period is longer).

Function	b value	Period
$y = \sin(3\theta)$		
$y = \sin(\frac{1}{2}\theta)$		
$y = \sin(4\theta)$		

Phase Shift (Horizontal Translation): $g(\theta) = \sin(\theta + c)$

Replacing θ with $\theta + c$ shifts the graph _____ by $|c|$ units. Phase shift = _____ (use the factored form to read it correctly).

Important — Factoring first! Before reading the phase shift, factor:

$$\sin(2\theta + \pi) = \sin(2(\theta + \underline{\hspace{1cm}}))$$

Phase shift = $-\underline{\hspace{1cm}} = \underline{\hspace{1cm}}$

LO 3.6.A — Part 3: The General Form

General Sinusoidal Function:

$$f(\theta) = a \sin(b(\theta + c)) + d$$

Parameter	Feature	Formula
a	Amplitude	$ a $ (sign determines reflection)
b	Period	$T = \frac{2\pi}{ b }$
c	Phase shift	$-c$ (left if $c > 0$, right if $c < 0$)
d	Midline	$y = d$

Worked Example: Identify all features of $f(\theta) = -2\sin\left(\frac{1}{2}(\theta - \pi)\right) + 3$.

Amplitude: _____

Period: _____

Phase shift: _____

Midline: _____

Reflected? _____

Sine–Cosine Equivalence (EK 3.6.A.1):

$$\cos \theta = \sin\left(\theta + \frac{\pi}{2}\right)$$

Any cosine function can be rewritten as a sine function with a phase shift.

Example: Rewrite $2\cos \theta$ as a sine function.

$$2\cos \theta = 2\sin(\theta + \text{_____})$$

Guided Practice

Guided Practice 1. For each function, identify the amplitude, period, phase shift, and midline.

Function	Amplitude	Period	Phase Shift	Midline
$f(\theta) = 5\sin(3\theta) - 2$				
$g(\theta) = -\cos\left(\frac{\pi}{2}\theta\right) + 4$				
$h(\theta) = 3\sin\left(2\left(\theta + \frac{\pi}{6}\right)\right)$				

Guided Practice 2. The function $k(\theta) = 4\sin(3\theta - \pi) + 1$.

- Factor the argument: $3\theta - \pi = 3(\theta - \text{_____})$
- Phase shift = _____ (left or right by how much?)
- List all four features: amplitude _____, period _____, phase shift _____, midline _____.